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Tunable nanophotonic devices and cavities based on a two-dimensional magnet

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Ahmet Kemal Demir [1,2], Luca Nessi [1,2], Sachin Vaidya [1], Connor A. Occhialini [1], Marin Soljačić [1] & Riccardo Comin [1]

Central to the field of nanophotonics is the ability to engineer the flow of light through nanoscale structures. These structures often have permanent working spectral ranges and optical properties that are fixed during fabrication. Quantum materials, with their correlated and intertwined degrees of freedom, offer a promising avenue for dynamically controlling photonic devices without altering their physical structure. Here we fabricate photonic crystal slabs from CrSBr, a van der Waals antiferromagnetic semiconductor, and demonstrate in situ control over their optical properties. Leveraging the combination of the exceptionally large refractive index of CrSBr near its excitonic resonances and its tunability via external fields, we achieve precise manipulation of photonic modes at near-visible and infrared wavelengths, showcasing a new paradigm for nanophotonic device design. The resulting guided resonances of the photonic crystal are tightly packed in the spectrum with very small mode volumes, are highly tunable via external magnetic fields and exhibit high Q factors exceeding 1,200. These resonances self-hybridize with the excitonic degrees of freedom, resulting in intrinsic strong light–matter coupling. Our findings underscore the potential of quantum materials for developing in situ tunable photonic elements and cavities.

Harnessing emerging quantum materials in photonics represents a promising research frontier. On the one hand, several efforts have been made to utilize these materials either as monolithic devices or as a part of a larger structure. Examples include guided mode resonances (GMRs) in atomically thin semiconductors[1], lasers[2] with ultrasmall active regions[3,4], ultrahigh-precision transition metal dichalcogenide (TMD) metasurfaces[5], atomically thin flat lenses[6] and mirrors[7,8], and hexagonal boron nitride (hBN) nanophotonic structures[9,10]—typically focusing on materials like TMDs and hBN rather than more exotic options like superconductors or quantum magnets. On the other hand, photonic cavities have been used to increase the interaction of matter modes with light. If the interaction strength is strong enough, light–matter hybrid modes (polaritons) can be formed, which have been highlighted by numerous fundamental studies of the light–matter coupling and their potential applications[11–16]. The appearance of polaritons in solids

光（电磁场）与物质（电子，声子、激子）相互作用，
导致能量在二者之间周期性交换，形成新量子物态

have been reported both with external cavities (Fabry–Pérot (FP)[17,18] and photonic crystal modes[12,19]) and with self-hybridized cavities[20–22] where the material itself provides the photon confinement in the structure. Cavity engineering has been shown to modify macroscopic properties in equilibrium via feedback on the effective temperature of the system[23] (thermal Purcell effect) and is predicted to affect the ground and excited states of many other quantum materials[24,25].

A promising but less explored aspect of quantum materials is that they can often be controlled on demand using external fields[26–28]. Photonic devices stand to benefit greatly from the incorporation of quantum materials, as the fabrication of photonic nanostructures generally does not allow in situ tunability of the resulting geometry once the design is set. Of great potential among quantum material-based photonic structures are monolithic devices where modes are able to fully reside in the material. This could translate the unique properties

[1]Department of Physics, Massachusetts Institute of Technology, Cambridge, MA, USA. [2]These authors contributed equally: Ahmet Kemal Demir, Luca Nessi. e-mail: rcomin@mit.edu

and tunability of quantum materials into the realm of nanophotonics while taking advantage of the maximum overlap of the photonic and matter modes for the strongest light–matter coupling. This is favourable over most common strong coupling examples with proximitized cavities where the coupling strength is necessarily reduced owing to the coupling occurring through evanescent fields only[21]. The realization of such monolithic devices is rare, probably due to a combination of insufficient characterization needed for device design and lack of established protocols for the nanofabrication.

In this work, we design and experimentally demonstrate in situ tunable nanophotonic modes in photonic crystal (PhC) slabs based on the low-dimensional magnet, CrSBr. We devise a special fabrication method to nanopattern CrSBr flakes and observe the formation of multiple high-Q GMRs and epsilon-below-zero modes that occupy small mode volumes. By applying external magnetic fields, we show how the mode of operation of the photonic crystal can switch from a typical elliptic mode to a hyperbolic one without changing the device geometry, driven by a massive change in the real permittivity ($\Delta\epsilon_1 \approx 800$ near the excitonic resonance).

This work makes two key contributions. We first show that the combination of nanophotonics and novel quantum materials like CrSBr leads to enhanced optical properties, particularly in terms of tunability and drastic miniaturization in a standalone, monolithic device. Second, we show that the photonic modes of the device self-hybridize with the matter modes, altering the optical properties of the sample through polariton formation. Both through the initial nanostructure design and through the in situ application of magnetic fields after fabrication, we observe tunable intrinsic strong coupling over a wide range of parameters including sample thicknesses, patterning design features and spectral ranges.

Our work establishes CrSBr as a uniquely powerful platform for tunable photonics, leveraging its exceptionally large and highly tunable refractive index, which surpasses conventional van der Waals materials such as hBN and TMDs. Unlike phase-change materials that usually offer only binary switching, CrSBr enables continuous refractive index switching across a wide spectral range, driven by its intertwined magnetic and optical properties. The strong coupling between the material's magnetic ground state and its optical response may allow rapid, reversible tuning using various external stimuli—including strain[29,30], pressure[31], ultrafast optical pulses[32] (by heating or magnon excitation) and both a.c.[33] and d.c. magnetic fields—and switching speeds as fast as 30 GHz, that is, the antiferromagnetic (AFM) magnon frequency. In addition, CrSBr supports high-quality simple photonic resonances, with experimentally observed Q factors exceeding 1,200, substantially enhanced by its large permittivity. This distinct combination of a giant refractive index, tunability through multiple external parameters and robust high-Q resonances makes CrSBr a suitable material for active photonic devices, outperforming traditional dielectrics and van der Waals materials. Our findings not only reveal a new degree of control in photonic crystal designs but also open pathways towards magnetically tunable photonic elements and exciton–polariton devices with unparalleled flexibility and performance.

Results

CrSBr is an air-stable van der Waals antiferromagnet that recently sparked interest due to its relatively high Néel temperature ($T_N = 132$ K) and the large anisotropy in its optical, magnetic and structural properties[34–37]. We first investigate the magnetic field dependence of the complex permittivity tensor in unpatterned CrSBr flakes. These flakes, exfoliated from bulk crystals onto sapphire substrates with thicknesses ranging from 10 to 300 nm, were systematically analysed using reflectivity spectra measurements with incident light polarized along the two principal crystallographic directions, a and b (ref. 38). The real part of the permittivity along the b axis is shown in Fig. 1b. We observe two main contributions: one excitonic resonance at around 910 nm and the other at 710 nm, which are labelled as X_1 and X_2, respectively. The optical properties of X_1 are well studied[27,37]. Recent work showed that X_2 is characterized by a similar oscillator strength and a more pronounced magnetic field dispersion, but also a reduced lifetime[38–40]. We note that the switching behaviour shown in Fig. 1b is highly robust, with all measured flakes exhibiting consistent refractive indices in both magnetic ground states. CrSBr flakes are exfoliated onto Si/SiO_2 (285 nm) substrates and subsequently patterned by reactive ion etching through silicon nitride masks to obtain photonic crystal slabs. These masks were previously defined by focused ion beam lithography and then transferred onto the CrSBr flakes using polymer-based dry-transfer techniques[41,42] (Methods and Supplementary Fig. 1). This fabrication method does not involve a lift-off or mask removal step; as a result, the flakes that are weakly bound to the substrate are not damaged or modified after fabrication. To retain the freedom to define an additional breaking of in-plane rotational symmetries in the fabrication design, we etch a periodic array of elliptical holes that can be rotated about an out-of-plane axis.

We measure reflectivity (Fig. 1c) and photoluminescence (PL) (Supplementary Fig. 2) from 30-nm-thick and 6-nm-thick patterned samples (see Fig. 1 for patterning parameters) at 5 K. Optimizing the parameters of the design geometry, we observe several GMRs, brightened by the periodic patterning, together with an FP mode determined solely by the flake thickness and not present in the 6-nm-thick flake. The permittivity tensor extracted from the experimental data and prior work are then used as input to rigorous coupled wave analysis (RCWA)[43] and finite element method solvers for reflectivity simulations. This accurately reproduces the observed spectral response as shown in Fig. 1c (for more details, see the Methods). The field profiles of these modes are calculated by finite element simulations and shown in Fig. 1d. As the real part of the refractive index in the 910–930 nm region is extremely large ($n = 6$–20), higher-order modes are uniquely very close to one another spectrally (two to three GMRs in $\Delta\lambda = 2$ nm for these patterning parameters). This contrasts sharply with, for example, silicon, where the same patterning would result in the order of hundreds of nanometres of spectral separation. We note that, due to the orders-of-magnitude difference in the in-plane and out-of-plane permittivities (Methods), the modes are strongly transverse electric (TE). As a result, we will refer to them as TE_{ml} (see the Methods for the convention we use). The TE_{41} resonance has a Q factor greater than 500, while the TE_{21} resonance Q factor is around 150. As a result of the anomalously large refractive index near the excitonic resonance, all photonic bands we report for the 30-nm-thick slab are essentially flat in momentum space (Supplementary Fig. 2). By the same token, the large refractive index is also responsible for the miniaturization of the device: the waveguide thickness is $\lambda/30$ from 30-nm-thick slabs, which can be enhanced as much as to $\lambda/150$ from GMRs of 6-nm-thick CrSBr on SiO_2 as shown in Fig. 1c and Supplementary Fig. 3.

The introduced patterns also act like monolithic photonic cavities and result in self-hybridization between the photonic mode (GMR) and the exciton (X_1). The resulting exciton–polaritons are bright in PL, and their energy can be tuned by external magnetic fields (Supplementary Fig. 2). These patterns allow us to explore light–matter coupling across a wide spectral region, without limitations imposed by slab thickness for FP mode formation[22,44]. The Rabi splitting due to TE_{21} in the 30-nm-thick sample is $\hbar\Omega_R = 66$ meV, an order of magnitude larger than $\hbar\Omega_R < 7$ meV from the strong coupling of the FP mode to the exciton (Supplementary Fig. 2).

We also note that for a large subset of patterns and thicknesses, we observe reflectivity dips in the epsilon-below-zero region ~860–908 nm (Fig. 1b), indicating the creation of absorption modes. The imprinted patterns generate bright PL peaks in the hyperbolic region, implying the formation of hyperbolic exciton–polaritons (HEP)[45] that are also tunable with magnetic fields (Supplementary Figs. 4, 5 and 16; also see Supplementary Note 1 and Supplementary Figs. 9–15).

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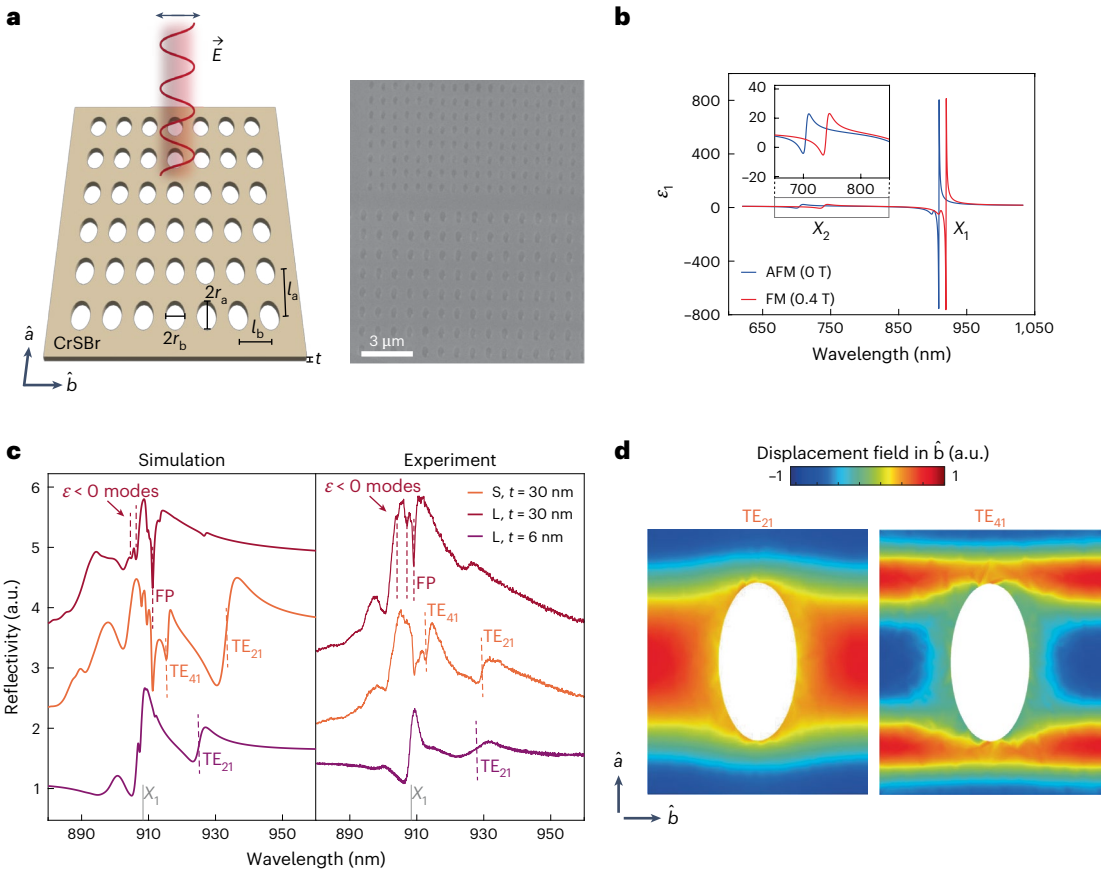

Fig. 1 | Photonic resonances in PhC slabs made of CrSBr. a, An illustration of the patterned CrSBr sample with a representative SEM image (top pattern is 'L' as given below). The light polarization is parallel to the *b* axis throughout the work. **b**, The permittivity of the *b* axis of unpatterned CrSBr at 5 K, extracted from thickness-dependent normal reflectivity. X_1 and X_2 are the two excitons defined in the text. Polarizing the spins of the material by applying a magnetic field of 0.4 T along the *b* axis switches the ground state from AFM to FM, modifying the permittivity substantially. **c**, Patterning the sample results in brightening the dark modes of the system, allowing the detection of the modes other than FP dips in reflectance. For the larger pattern, 'L', (red curve) (r_a = 180 nm, r_b = 90 nm, l_a = 600 nm, l_b = 500 nm), discernible modes appear primarily in the epsilon-below-zero region for the 30-nm-thick flake. A TE-like mode is visible for the 6-nm-thick flake (Supplementary Fig. 3). For the smaller pattern, 'S', (orange curve) (r_a = 120 nm, r_b = 60 nm, l_a = 400 nm, l_b = 333 nm, t = 30 nm), a number of TE-like modes appear, in good agreement with the RCWA simulations. The feature around 926 nm in the experimental data is an intrinsic material peak, not a photonic mode (Supplementary Fig. 2d). **d**, Field profiles of the GMRs of 'S' sample in **c** from FEM simulation.

We demonstrate the creation of GMRs in another, thinner (thickness (t) = 20 nm) sample (Fig. 2). Patterns, with size progression, are transferred onto a large, uniform flake. For the first experiment, the ellipses are patterned with a 20° rotation around the out-of-plane axis as shown in the scanning electron microscopy (SEM) image of Fig. 2a. We again collect reflectance and PL and observe the spectral progression of the coupled waveguide mode with respect to the varying lattice parameters. The patterning weakens the FP mode in the highest index region and brightens the waveguide modes. All features in the epsilon-positive region in the spectrum can be ascribed to GMRs, according to the simulations. The dips observed for the semi-major axis r_a = 90 nm and r_a = 60 nm patterns are wider than the ones observed in Fig. 1, as there are two GMRs that are intriguingly just 3 nm apart due to the real permittivity jumping from 100 to 450 when going from 915 nm to 910 nm in wavelength. This remarkable increase in permittivity pushes the waveguide modes closer to each other. Distinguishing these two modes is challenging even with normal-incidence reflectivity simulations (Fig. 2c), which agree well with the experiments. We identify these modes as TE_{41} and TE_{21} (Methods; the mode profiles are displayed in Supplementary Fig. 6). We also note that the PL maximum (Fig. 2b) follows the reflectivity dip in Fig. 2c, indicating the strong coupling of the exciton X_1 to the GMR cavity modes. We similarly observe PL emission emerging from the epsilon-below-zero region (below 908 nm) that gets more pronounced with the increasing absorption as shown in the reflectivity plot (Fig. 2b). We ascribe this absorption edge, again, to HEPs[45] that appear due to patterning.

In the second experiment, we fabricate one-dimensional (1D) grating-like structures from CrSBr flakes with thicknesses ranging from approximately 14 nm to 20 nm, as shown in Fig. 2d. Similar to our previous observations, we detect multiple GMR modes, with the Q factor of the TE_{41} mode reaching more than 1,200 in these ultrathin (~λ/50) photonic crystal slabs. The observed reflectivity features and the mode evolution with varying thickness align closely with our simulations (Fig. 2e). It is important to highlight that the high-Q nature of these modes does not stem from any specialized photonic design aimed at increasing Q factors—these are standard GMRs without symmetry-based decoupling from the radiation continuum (also see Supplementary Note 2). Instead, the exceptionally high Q factors arise from the intrinsically large refractive index of CrSBr, which compresses what would otherwise be a broad photonic mode if the same structure were made from a material like hBN. This makes the high-Q modes in CrSBr fundamentally unique, as they emerge as a direct consequence of the material's inherent optical properties rather than deliberate photonic engineering.

The unique properties of CrSBr arise from both its exceptionally large and anisotropic refractive index and the dependence of these optical properties on its ground state. As shown in Fig. 1b, the 910-nm

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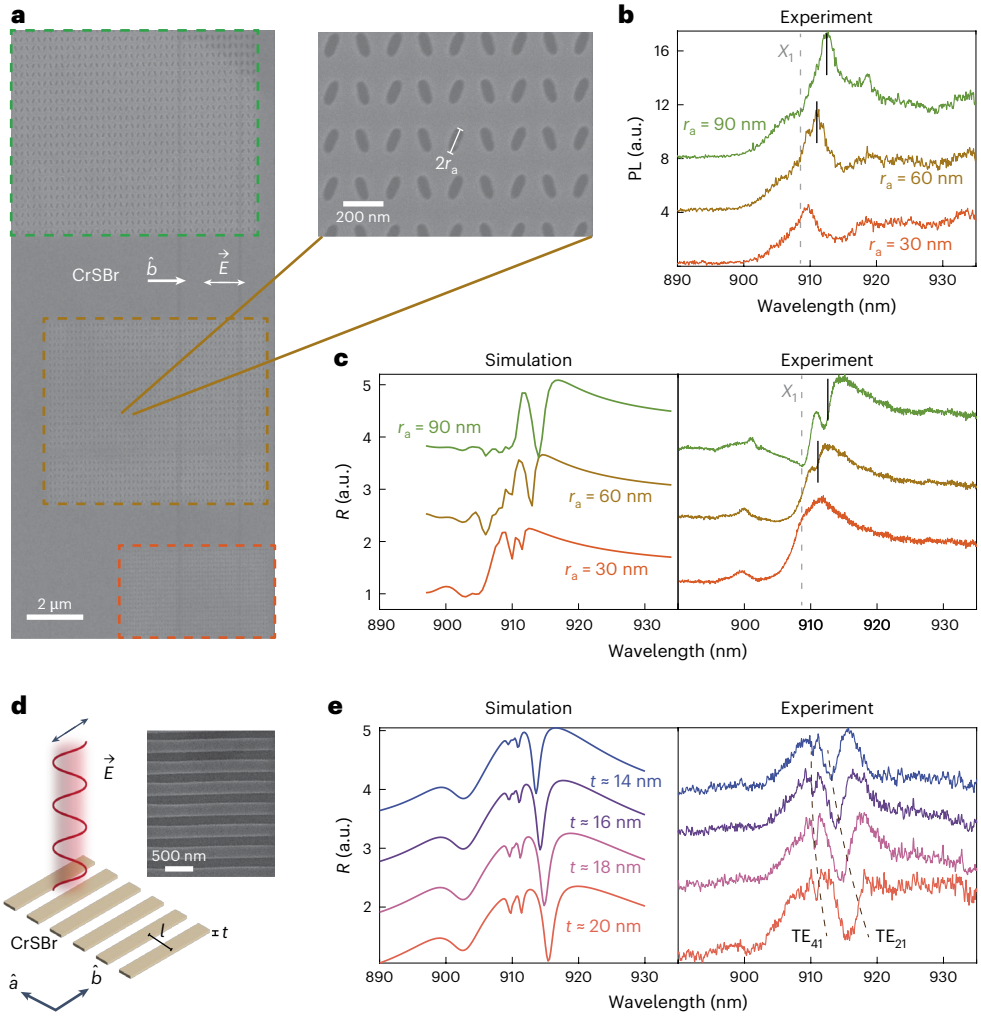

Fig. 2 | Self-hybridized exciton–polaritons in CrSBr photonic crystal slabs. **a**, A SEM image of the sample. Ellipses are rotated by 20° for this sample, as shown. The colour code is as in **b**. **b**, PL from three samples with different patterning parameters. Green: $r_a = 90$ nm, $r_b = 40$ nm, $l_a = 300$ nm, $l_b = 200$ nm, $t = 20$ nm; brown: $r_a = 60$ nm, $r_b = 26$ nm, $l_a = 232$ nm, $l_b = 157$ nm, $t = 20$ nm; orange: $r_a = 30$ nm, $r_b = 14$ nm, $l_a = 150$ nm, $l_b = 100$ nm, $t = 20$ nm. The lower energy part of the PL spectrum has a fine structure probably due to exciton–phonon interactions and is out of scope of this study. We note the finite PL from the epsilon-below-zero region. **c**, The reflectivity spectrum of the samples introduced in **b**. Guided resonance maxima follow the PL maxima for all samples, indicating strong coupling of the photonic mode to the exciton. The straight line indicates the GMRs TE_{21} and TE_{41}, and the dashed line indicates the centre of X_1. We also note the developing dip around 905 nm, indicating allowed absorption modes in the epsilon-below-zero region. **d**, A schematic of the ultrathin 1D CrSBr photonic crystal slab, along with an SEM image of the structure. $l = 400$ nm; area filling factor is 50%. **e**, Simulations and experiments on the 1D photonic crystals described. The Q factors for the TE_{41} mode for $t = 18$, 20 nm slabs are as large as 1,200.

X_1 exciton absorption centre redshifts by as much as 10 nm (15 meV) when the magnetic ground state of the material is switched from AFM to ferromagnetic (FM) when the spins are polarized[27]. Remarkably, the magnetic field required to polarize the spins is also very anisotropic. If the magnetic field is applied parallel to the b axis (the easy axis) of the material, the spins flip abruptly at around 0.34 T. For the a and c axes, the spins rearrange continuously, and the saturation occurs at around 1 T and 2.4 T, respectively.

We leverage the sensitivity of CrSBr to its environment to tune the photonic resonances we reported above in this work in a reversible manner. We apply a relatively small magnetic field of 0.4 T parallel to the b axis of the CrSBr sample shown in Fig. 2. The AFM-to-FM spin flip transition is accompanied by the rigid redshift of the photonic resonances by as much as $\Delta\lambda = 10$ nm. This implies that, upon the application of the magnetic field, modes residing around $\lambda = 91$ nm switch from GMRs in the positive-permittivity region to HEP resonances (Fig. 3a). Therefore, on-demand operational mode switching is achieved in a photonic crystal without altering its geometry.

When the magnetic field is turned off, the resonances return to their original position (however, see Supplementary Fig. 17 for an attempt to introduce non-volatility). It is also possible to tune the spectral position of the resonances continuously, by applying a magnetic field in any perpendicular direction (a or c axes). We exemplify this in Supplementary Fig. 4 by shifting the PL emission from the HEPs continuously through the application of magnetic fields parallel to the a axis of the crystal. As the origin of the transition is electronic[27], the mode shifting is robust without any limitations on the number of switching operations.

The b-axis permittivity of CrSBr also depends on the temperature. As the temperature increases from 5 K to 55 K, the exciton absorption peak redshifts and broadens. This redshift and broadening also affect the resonances owing to a weakening of the permittivity contrast (Fig. 3b).

We illustrate an even stronger tuning of these photonic modes with magnetic fields by utilizing the higher-energy exciton, X_2. This part of the optical spectrum of CrSBr is less well studied compared

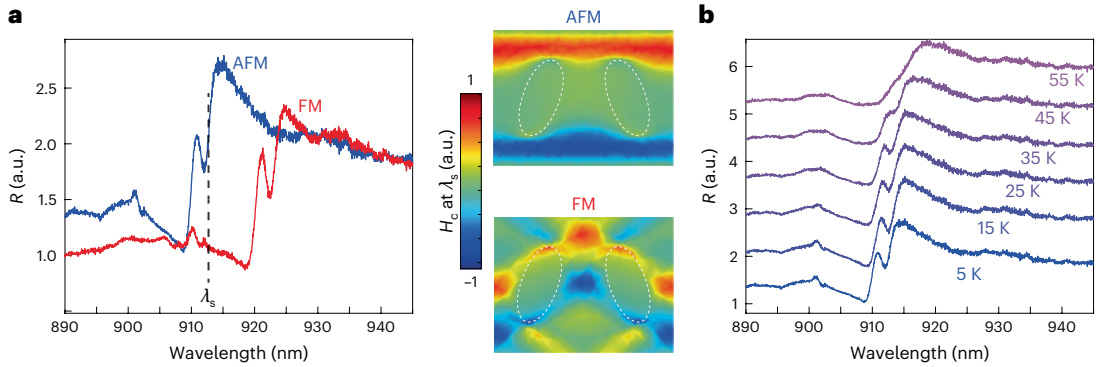

Fig. 3 | In situ tunability of the photonic modes. a, When CrSBr is magnetically polarized, the exciton energy (hence, the refractive index) redshifts. The photonic modes redshift accordingly. Here, the GMR of Fig. 2 shifts by 10 nm (15 meV) when a magnetic field of 0.4 T is applied along the *b* axis. The mode switching is illustrated by the mode profiles (magnetic field in the out-of-plane axis, H_c) at a given wavelength of operation. See Supplementary Note 1 for a discussion of the mode symmetries and Supplementary Fig. 6 for in-plane electric field mode profiles of the brightened HEPs. **b**, The permittivity depends on the temperature. Increasing the temperature results in a less pronounced permittivity anisotropy, weakening the spectral features.

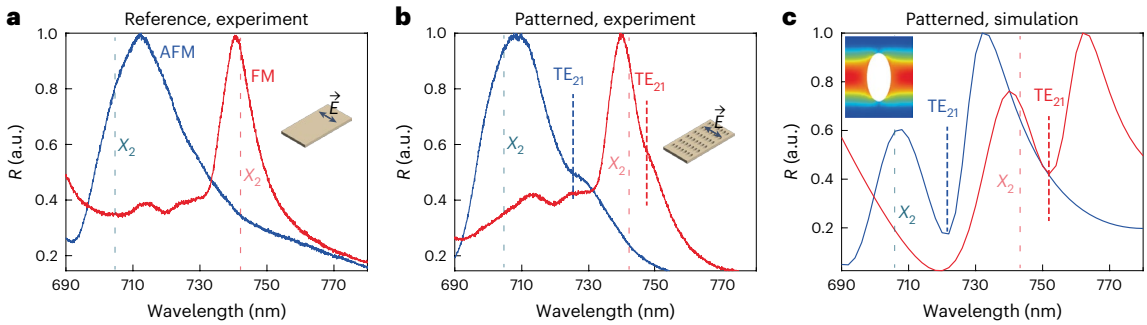

Fig. 4 | Strongly tunable photonic resonances in the near-visible range. a, Reference reflectivity spectra from unpatterned CrSBr, $t = 30$ nm. The permittivity peak shifts by 30 nm (70 meV) due to the shift of X_2 in a magnetic field. Approximate spectral positions of X_2 for AFM and FM states are indicated by faint dashed lines. **b**, Patterned samples host GMRs that are magnetic field tunable by 25 nm (50 meV). Patterning parameters: $r_a = 120$ nm, $r_b = 60$ nm, $l_a = 400$ nm, $l_b = 333$ nm, $t = 30$ nm. **c**, Normal reflectance simulations of the patterned sample agree with the observed modes. Inset: displacement field profile in the *b* axis for the GMR mode (TE_{21} in the AFM phase).

with X_1, probably due to a combination of weaker PL emission and higher losses around the centre of exciton absorption. Nonetheless, the magnetic field dispersion of X_2 is almost an order of magnitude stronger than X_1 in terms of absolute energy shift, as shown in the permittivity data (Fig. 1b) and in a reference reflectivity measurement on an unpatterned flake (Fig. 4a). This characteristic makes the X_2 resonance more interesting for in situ tunability purposes and to form GMRs in the near-visible range. These comparisons underscore that CrSBr offers a high level of tunability, far exceeding what has been demonstrated in TMD-based systems[46] (see Supplementary Table 1 for phase-change materials' comparison with CrSBr). While TMD cavities show spectral shifts of just 1 meV under 5 T (0.2 meV T^{-1}) and 0.5 nm via electrical gating, CrSBr exhibits shifts of ~50 meV T^{-1} along the *a* axis and discrete 50 meV tuning along the *b* axis under <0.4 T, with total spectral shifts exceeding 20 nm. This represents an enhancement of over two orders of magnitude compared with TMDs. The ability to achieve such dramatic spectral shifts with relatively small magnetic fields highlights the unique magneto-optical coupling in CrSBr, enabling highly efficient and widely tunable photonic devices that are fundamentally inaccessible in conventional TMD systems.

We perform reflectivity and PL measurements on a patterned slab of CrSBr with a thickness of 30 nm (Fig. 4a,b and Supplementary Fig. 8). Similar to GMRs near X_1, we observe a reflectivity dip in the high-permittivity region (Fig. 4b). Upon application of the magnetic field, the spins are polarized along the *b* axis while the GMR mode redshifts as much as 25 nm (50 meV), or three times more than X_1.

It should be noted that the reflectivity dip is much wider than the X_1 resonance. This is expected, as the permittivity is an order of magnitude lower compared with the region of interest around X_1. This results in photonic bands being relatively more dispersive in momentum space. Even without momentum dispersion and with normal-incidence reflection, the modes exhibit lower *Q* factors (Fig. 4c) than near-X_1 GMRs, due to increased losses and reduced index contrast.

Discussion

Altogether, we demonstrate ultrathin (as thin as $\lambda/150$) CrSBr photonic crystal slabs supporting in situ magnetically tunable resonant modes. These guided modes self-hybridize with excitons, forming exciton–polaritons controllable by both structural design and magnetic fields. The findings of this study may be advanced through modifications to the design of photonic crystals or by utilizing different materials in their fabrication. In addition, the potential for spectral tuning can be harnessed across a substantially broader spectral range.

We opted to work mainly within the thickness range 6–30 nm for the PhC slabs. However, thinner PhC slabs, down to the monolayer limit, are feasible, although perhaps with a lower permittivity environment. These PhC slabs offer a direct pathway to higher *Q* factors through bound states in the continuum (BIC), enabling stronger light–matter interactions and lasing[47–51].

The focus of this work is monolithic optical nanostructures made of CrSBr operating at cryogenic temperatures, which are necessary because the magnetic order sets in only below 132 K. Nevertheless,

we stress that the design principles we put forward in this work are not specific to CrSBr. Recently studied van der Waals materials are showing promising properties that can be utilized in the realization of nanophotonics structures. This includes other materials similar in structure (for example, CrOCl, ref. 52) or in optical properties (for example, $CrPS_4$, ref. 53) that can work also in different spectral ranges (for example, $NbOCl_2$, refs. 54,55). Moreover, other tuning knobs can be utilized by exploiting other phase transitions, such as ferroelectricity (for example, $NbOCl_2$) or multiferroicity (for example, NiI_2, ref. 56) adding the possibility to use other external fields such as electric fields[57], strain or pressure. We note that the optical properties of CrSBr photonic structures can also be tuned by gating[57].

Although the redshift of the exciton wavelengths across the magnetic transition is relatively small (10 nm and 40 nm for X_1 and X_2, respectively), due to the extremely large oscillator strength of the excitons, the effect of the magnetic fields is not limited to the redshift region. The permittivity contrast due to the transition from AFM to FM (defined as the absolute value of $\Delta\epsilon_B = \epsilon_{1,AFM} - \epsilon_{1,FM}$) is larger than the permittivity contrast between silicon nitride, a widely used semiconductor in photonics, and vacuum in a wavelength range of over 180 nm across the visible and near-infrared spectrum (Supplementary Fig. 7). A direct implication is the use of CrSBr slabs, even without the strict need for patterning, as magnetic-field activated switches to alter the dielectric medium for other optical components. Through periodic patterning, the effective permittivity of the slabs is modified strongly due to the waveguide modes being folded onto the light cone. This mechanism enables the enhancement or suppression of the permittivity contrast $\Delta\epsilon_B$ where the guided modes appear, facilitating precise spectral tunability in the far field. This feature makes the PhC slabs discussed in this work ideal for tunable optical filters and modulators that can also be extremely thin and operate at a wide angle (Supplementary Fig. 2), due to the unique combination of high refractive index and tunability.

Due to the anisotropy of CrSBr, the refractive index change induced in the flakes upon the application of a magnetic field depends on their relative orientation, such as when they are twisted on top of one another. This twist-dependent variation enables precise control of optical properties, which can support tunable exotic optical modes. More importantly, the van der Waals nature of CrSBr allows easy integration with other systems without requiring lattice matching. This, combined with the ability to spatially modulate the refractive index in situ, unlocks a compelling route to modify a photonic system's topological invariants. Such modifications can be realized through small d.c. magnetic fields or even via all-optical methods, offering dynamic and reconfigurable control over topological photonic phases[11,58–63].

Online content

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Methods

CrSBr crystals

All crystals used in this study are purchased from HQ Graphene. The flakes are obtained by mechanical exfoliation from bulk single crsytals onto Si/SiO_2 or sapphire substrates. The approximate thickness of unpatterned flakes can be determined by a combination of apparent colour and reflectivity (for tracking the number and positions of FP modes according to the permittivity fits, see 'Permittivity fits' section). For more precise thickness information, we use atomic force microscopy.

Nanofabrication

Suspended silicon nitride membranes are purchased from NORCADA (TA301X, 50 nm thick). We used the Raith VELION FIB-SEM, at Characterization.nano of MIT.nano, for the lithographic patterning of the membranes. After inspecting the patterns (with SEM or optical micrography), we transfer these membranes on the sample of choice using well-established polymer-stamp transfer techniques. One technique we find particularly simple is attaching a small piece of polydimethylsiloxane on a glass stamp and picking up the membranes directly from the silicon support. The pick-up success is significantly higher when the membranes are first annealed at 150 °C for 10–30 min and when polydimethylsiloxane is lifted off very fast after the contact with the cooled-down membrane. Another method we used is to mill the edges of the membrane on a flat substrate with focused ion beams (FIB) and then pick up the membrane using the transfer technique described above. After the transfer, we perform reactive ion etching (Samco Plasma Etching System RIE-230iP) to etch away the parts under the patterns in SiN membranes. We use a chlorine recipe, normally used to etch chromium, to etch CrSBr: Cl_2 20 sccm and O_2 3 sccm. This recipe results in fast etching rates (up to 10 nm s^{-1}) for CrSBr, while SiN is minimally etched. See Supplementary Fig. 1 for an illustration of the fabrication steps. Another, more general, method would be to use argon milling. We confirm that CrSBr is completely etched first by optical images (CrSBr flakes under SiN appear the same colour while the unprotected regions disappear) and later by SEM, where the lack of CrSBr results in a larger contrast especially for thicker samples. We note that we do not remove the SiN membranes after the fabrication; they are incorporated into the structure. As SiN has a very small relative permittivity compared with CrSBr in the spectral regions we are reporting (4 versus 100–450 for X_1, 4 versus ~20 for X_2), leaving the membrane on CrSBr after the patterning is inconsequential.

Optical spectroscopy

We use a supercontinuum light source (NKT Photonics SuperK Fianum) as the excitation source for reflectivity measurements. Broadband light is first polarized by a polarizer and an achromatic half-wave plate to the *b* axis of the crystal and then focused on the sample by a 20× objective (numerical aperture 0.25) in a cryostat (Montana S50) equipped with a 1 T in-plane magnet. All measurements are performed at 5 K unless otherwise noted. Using an objective without back-focal plane imaging would normally be detrimental to the observation of dispersive features in the spectrum because of the momentum spread. However, in this case, the in-plane refractive index is so large that the photonic bands arising from coupling with the excitons are flat, essentially eliminating the need for angle-resolved spectroscopy (Supplementary Fig. 5). The collected light from the sample is then sent to a spectrometer (Horiba LabRAM Evolution). PL measurements use the same set-up with a 532-nm CW laser instead of the supercontinuum laser. All reflectivity spectra near the first exciton presented in this work are normalized by the broadband reflectance from the SiN/285-nm-SiO_2/Si interface ($R \approx 0.35$ around 900 nm). This is not possible for spectra around the second exciton (Fig. 4) owing to thin film interference minimum of the substrate; so, we opted to normalize to 1.

Permittivity fits

We performed reflectivity measurements on a number of unpatterned CrSBr flakes with varying thickness on sapphire[38]. Due to the extremely large refractive index around X_1 and X_2, we start observing sequential FP modes with increasing thickness. We fit a Lorentzian for all the features visible in the reflectivity spectrum for the *b* axis as follows:

$$\epsilon(E) = \epsilon_b + \sum_i \frac{f_i}{E_i^2 - E^2 + iE\Gamma_i},$$

where E is energy, ϵ_b is the background permittivity (which we fit as 11.1, similar to refs. 44,45), f_i is the oscillator strength, E_i is the energy of the exciton and Γ_i is damping for the oscillator. We feed these variables into a transfer matrix method solver as the permittivity of a slab with a given thickness on sapphire substrate. We measure 16 thin flakes with different thicknesses, ranging from 13 nm to 208 nm, and fit the reflectivity data from the transfer matrix method to the experimental data. This is relatively straightforward, as even the flakes as thin as 60 nm exhibit multiple FP dips. See ref. 38 for the reflectivity data (with only two oscillators fit). For this work, we also fit a third oscillator, X^*, as we are also interested in the hyperbolic permittivity region. The permittivity values given in Fig. 1b are obtained from the following parameters:

$$X_1 : f_{X_1} = 1.7(\text{eV})^2,\ E_{1,\text{AFM}} = 1.3637\ \text{eV},\ E_{1,\text{FM}} = 1.3481\ \text{eV},\ \Gamma_1 = 0.0008\ \text{eV}.$$

$$X_2 : f_{X_2} = 1.2(\text{eV})^2, E_{2,\text{AFM}} = 1.76\ \text{eV},\ E_{2,\text{FM}} = 1.67\ \text{eV},\ \Gamma_2 = 0.025\ \text{eV}.$$

$$X^* : f_{X*} = 0.5(\text{eV})^2,\ E_{X*,\text{AFM}} = 1.3771\ \text{eV},\ E_{X*,\text{FM}} = 1.361\ \text{eV},\ \Gamma_{X*} = 0.0075\ \text{eV}$$

We take the *a*-axis and *c*-axis relative permittivity values as 11 and 4 from prior studies[44,45]. These permittivity values are then used for both finite element analysis (FEM) and RCWA simulations.

Simulations

We use S4 (ref. 43) for RCWA (Fig. 1c) and COMSOL Multiphysics for FEM simulations (Figs. 1d, 2c and 4d), using the permittivity values given above, and the geometry described in the main text: air/patterned SiN/patterned CrSBr with a given thickness/285-nm-thick SiO_2/Si. All simulations presented in the main text are normal-incidence reflectance; we present angular dependence in Supplementary Figs. 2 and 5. TE_{ml} is defined as TE modes that have *m* antinodes in the electric field profile in the *a*-axis and *l* antinodes in the *b* axis. We deduce this from the mode profiles FEM yields. We again note that we call TE-like modes TE in the main text. In a photonic crystal slab, there are no pure TE or TM modes. All simulations are normalized such that R is between 0 and 1 except for Fig. 1c, where we scaled the R by the reference reflectance from the substrate (0.35) so that the experimental and theoretical values are directly comparable.

Data availability

Raw experimental data used in the article are available via Harvard Dataverse at https://doi.org/10.7910/DVN/1NPBTC.

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Author contributions

A.K.D., L.N. and R.C. conceived the project. A.K.D. performed the nanofabrication and carried out the experiments together with L.N., with help from C.A.O., under the supervision of R.C. S.V., supervised by M.S., and A.K.D. performed the simulations. All authors contributed to the writing of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Riccardo Comin.

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